Name of the module: **bitRev256**
Target delivery date: 12-23-16
Location in GIT repository: vspa_common\vspa-lib\bitRev\ bitRev256

Type of code:

IPPU assembly function (1 function)
C function invoking IPPU (1 function)

IPPU Function API:

```
extern void bitRev256(void);
```

C function invoking IPPU function API:

```
extern bool bitRev256Invoke(
    vspa_complex_float16        *pOut,
    vspa_complex_float16 const  *pIn
);
```

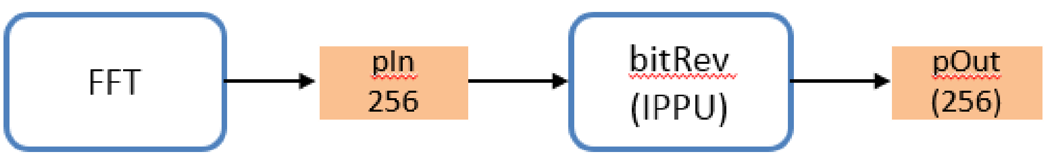
Return value: True when IPPU is not busy, false when IPPU is busy

API description:

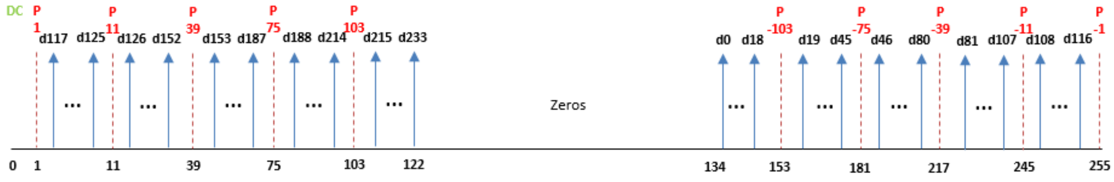
bitRev module performs shifting and bit-reversal on the input buffer of 256 sub-carriers.

- pOut pointer to linear ordered sub-carriers buffer in VCPU. (dmem aligned)
- pIn pointer to the input buffer in VCPU dmem containing sub-carriers in bit-reversed order. (32 bit aligned)

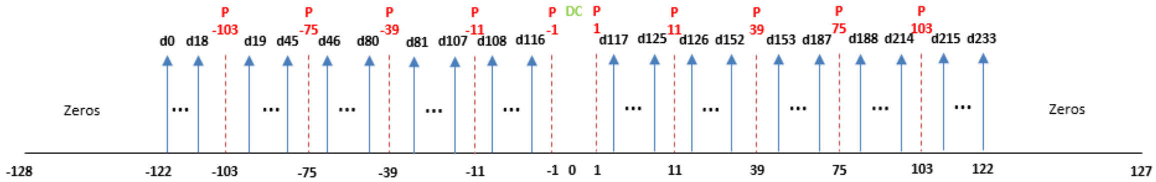
Implementation Plan:



Here is the sub-carrier arrangement in output buffer WITHOUT bit-reversal addressing(Note: the module expects the following sub-carriers in bit-reversed address)



For the above input(in bit-reversed order) the output looks like the following



DMEM requirements:

Variable	Minimum size (half-words)	Minimum alignment (half-words)	Allocated by
pOut	512	64	Caller
pIn	512	2	Caller

Test Plan:

Verify the bitRev output with matlab reference for the following use-cases

- 802.11ac 80 MHz